Introducing practice quizzes in organic chemistry: a quasi-experimental comparison of two cohorts

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Abstract

We compared the 2019 cohort of an organic chemistry course, taught without quizzes, with the 2020 cohort, taught with weekly practice quizzes. Across 317 students the pass rate rose from 71% to 79%.

1. Methods

Design: quasi-experimental comparison of two consecutive cohorts (non-randomised, historical control). Participants: 317 students (2019 cohort, n = 152; 2020 cohort, n = 165). Intervention: weekly practice quizzes from 2020 onwards. Outcome: course pass rate.

2. Results

The pass rate was 71% in the 2019 cohort and 79% in the 2020 cohort (difference 8 percentage points, 95% CI -1 to 17).

3. Limitations

The cohorts were not equivalent at baseline, and the 2020 cohort moved to online teaching in the middle of the semester, which may explain part of the difference.

Fictional paper generated to test evidence-table extraction. Authors, journal, data and results are invented.